STRATEGIC ASSESSMENT REPORT • TRANSBOUNDARY GOVERNANCE & SECURITY

Climate and Water Diplomacy

Engineering the Atmosphere and Damming the Rivers

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1. EXECUTIVE SUMMARY

The 21st century marks a fundamental shift in international relations, characterized by the convergence of climate destabilization and unilateral technological capability. As traditional mitigation efforts struggle to keep pace with warming trajectories, sovereign states and private entities are increasingly turning toward dual mechanisms of environmental modification: **atmospheric geoengineering** (such as Solar Radiation Management and Carbon Dioxide Removal) and **macro-hydrological control** (such as transboundary mega-dams and river diversions). While framed as technical remedies for resource scarcity or warming, these localized interventions generate severe transboundary security risks, asymmetric power dynamics, and unprecedented legal vacuums.

The central dilemma of modern climate diplomacy lies in the tension between unilateral execution and multilateral governance. A single nation now possesses the technological and financial leverage to alter global precipitation patterns through atmospheric aerosol injection, or manipulate downstream freshwater access through upstream dam operations. Without binding global treaty structures, these physical interventions risk triggering severe interstate conflict, ecological destabilization, and deep trust deficits across international borders.

263	0	1.5°C	60%+
TRANSBOUNDARY BASINS	BINDING SRM TREATIES	WARMING THRESHOLD	GLOBAL WATER FLOWS

STRATEGIC IMPERATIVE FOR MULTILATERAL GOVERNANCE

Traditional international environmental law—built on reactive consensus—is structurally ill-equipped to govern unilateral climate interventions and hydro-hegemony. Establishing anticipatory frameworks, legal accountability mechanisms, and equitable benefit-sharing models is no longer merely an ecological preference, but a non-negotiable imperative for international peace and security.

2. THE CLIMATE ENGINEERING DILEMMA: SRM VS. CDR

As global emissions continue to overshoot targets established under the Paris Agreement, climate intervention technologies have transitioned from fringe theoretical concepts to high-stakes national security considerations. Climate engineering interventions are broadly categorized into two distinct technological pathways: **Carbon Dioxide Removal (CDR)** and **Solar Radiation Management (SRM)**.

Carbon Dioxide Removal (CDR) comprises techniques designed to extract greenhouse gases directly from the atmosphere and sequester them permanently in geological, oceanic, or terrestrial sinks. Key methodologies include Direct Air Capture (DAC), Bioenergy with Carbon Capture and Storage (BECCS), and Ocean Alkalinity Enhancement (OAE).

Because CDR directly addresses the root chemical cause of anthropogenic climate change, it enjoys relatively broad international support and political legitimacy. However, its primary constraints are temporal, economic, and logistical: CDR technologies remain prohibitively expensive, require massive energy inputs, and operate on slow, decadal scales that cannot immediately head off acute planetary tipping points.

In contrast, **Solar Radiation Management (SRM)** seeks to modify the Earth's radiative balance by reflecting a small percentage of incoming solar radiation back into space. Dominant approaches include Stratospheric Aerosol Injection (SAI)—which disperses light-scattering aerosols such as sulfur dioxide into the upper atmosphere—and Marine Cloud Brightening (MCB). SRM offers a rapid, highly potent, and surprisingly inexpensive method to decrease global mean temperatures within months. However, SRM does not reduce carbon concentrations, leaves ocean acidification unchecked, and carries extreme, unpredictable transboundary risks—most notably the systemic disruption of tropical monsoon cycles, altered agricultural productivity, and ozone layer degradation.

Dimension	Carbon Dioxide Removal (CDR)	Solar Radiation Management (SRM)
Primary Mechanism	Atmospheric CO_2 extraction and permanent sequestration.	Solar radiation reflection via stratospheric aerosol dispersion.
Speed & Response	Decadal to centennial timeline for measurable thermal cooling.	Immediate cooling impact observed within months of deployment.
Economic Cost	High deployment cost ($100–$600 per ton of removed CO_2).	Low direct deployment cost ($10B–$20B annually worldwide).
Transboundary Risk	Low global risk; localized land, energy, and ecosystem pressure.	Extreme global risk; altered monsoons, regional drought, termination shock.
Feasibility & Politics	Broad policy consensus; constrained by capital and physical scale.	Severe diplomatic flashpoint; extreme risk of unilateral deployment.

3. THE GOVERNANCE VACUUM & MORAL HAZARD DILEMMA

The Legal Gray Zone and Unilateral Risk

International law currently lacks a comprehensive, legally binding convention to explicitly regulate or prohibit atmospheric solar intervention. The **1976 UN Environmental Modification Convention (ENMOD)** prohibits the hostile or military manipulation of environmental processes, but fails to account for peaceful, defensive, or mitigation-oriented geoengineering interventions.

Similarly, non-binding decisions under the Convention on Biological Diversity (CBD) urge caution but lack mechanism for enforcement. This institutional void creates a dangerous exposure to "rogue deployment"—where a single climate-distressed state or wealthy private individual acts unilaterally to deploy SAI, imposing unconsented global environmental consequences.

Moral Hazard and the Global South Perspective

Beyond legal vulnerabilities, geoengineering creates a profound **moral hazard**: the prospect of a cheap technological "quick fix" diminishes political will among high-emitting industrial nations to sustain necessary emission reductions and fulfill international climate finance commitments.

Diplomats from the Global South vehemently oppose SRM research and deployment, characterizing it as a mechanism that allows high-income nations to externalize ecological risks. Developing nations—who contribute least to cumulative emissions—face the greatest threat of catastrophic monsoon shifts and agricultural failure caused by uncoordinated atmospheric management.

4. CASE STUDY: CANCELLATION OF HARVARD'S SCOPEX (2023)

The cancellation of Harvard University's **Stratospheric Controlled Perturbation Experiment (SCoPEx)** stands as a key historical indicator of the political and ethical resistance surrounding geoengineering testing. Originally designed to evaluate the physical behavior of aerosol plumes in the stratosphere using high-altitude balloons, SCoPEx initially sought field testing in Sweden in 2021. However, intense opposition from the Indigenous Saami Council and environmental organizations forced the project's suspension, driving researchers to explore alternate international partnerships, including discussions in Kenya in 2023.

The proposed relocation to Kenya triggered an immediate backlash across African civil society coalitions, climate justice advocates, and regional governance groups. Opponents leveled explicit charges of **"climate colonialism,"** arguing that Northern academic and financial institutions were exporting environmental risks to vulnerable Southern populations without formal multilateral oversight or genuine prior informed consent.

Faced with widespread social condemnation, the withdrawal of institutional backing, and mounting diplomatic scrutiny, project leaders officially terminated SCoPEx in late 2023. The project's failure sent an unmistakable signal to the international scientific community: technological research operating outside transparent, inclusive, and democratically sanctioned global governance frameworks will be met with decisive diplomatic resistance and public shutdown.

5. HISTORIC AND MODERN CLOUD SEEDING: SUCCESS, FAILURE, & CONFLICT

While solar radiation intervention remains largely theoretical, **tropospheric weather modification**—specifically cloud seeding via silver iodide, potassium iodide, or liquid propane—has operated commercially for decades. Modern programs deploy ground generators and aircraft to inject aerosol agents into storm systems to stimulate ice crystal formation, boosting precipitation, dispersing fog, or suppressing hail.

The practical implementation of cloud seeding demonstrates a distinct divergence between localized, controlled operational successes and severe uncoordinated hydrological failures:

- **Targeted Successes:** Orographic cloud seeding across snowpack regions (such as the US Sierra Nevada and Australian Snowy Mountains) has consistently yielded measurable 5% to 15% increases in annual snowfall, bolstering municipal runoff and hydroelectric power generation.
- **Operational Failures:** Uncoordinated cloud seeding in warm, hyper-arid urban environments frequently fails to yield rain or generates extreme cloudbursts and flash flooding that overwhelm drainage infrastructure, as observed in recent Middle Eastern storm events.

At the international level, weather modification is increasingly becoming a flashpoint for interstate friction. Neighboring states frequently exchange allegations of **"rain theft"** or atmospheric moisture interception. International water law offers no established mechanism to resolve claims regarding sovereign ownership over atmospheric vapor, leaving a dangerous regulatory gap as weather modification expands globally.

6. WATER DIPLOMACY CASE STUDY 1: CHINA'S UPSTREAM DAM BUILDING

Transboundary hydro-politics represents a physical manifestation of environmental power projection. The People's Republic of China occupies a commanding position as Asia's primary hydro-hegemon, holding territorial control over the Tibetan Plateau—the glacial source of the continent's major transboundary rivers, including the Mekong (Lancang), Brahmaputra (Yarlung Zangbo), Indus, and Salween.

China’s construction of mega-dam cascades—most notably along the Upper Lancang—grants Beijing unprecedented unilateral control over downstream river flow volumes, seasonal timing, and sediment distribution. Downstream nations across Southeast Asia (Thailand, Laos, Cambodia, Vietnam) face severe ecological disruption, riverbed erosion, and fishery collapse during dry seasons. Operating outside full membership in the downstream Mekong River Commission, China effectively leverages its hydro-infrastructure as a powerful strategic tool in bilateral diplomacy.

7. WATER DIPLOMACY CASE STUDY 2: THE INDUS WATER TREATY CRISIS

The 1960 **Indus Waters Treaty (IWT)** between India and Pakistan, mediated by the World Bank, has long served as a benchmark for durable transboundary hydro-diplomacy, surviving multiple armed conflicts between the two nuclear-armed neighbors. The treaty allocates the Eastern Rivers (Sutlej, Beas, Ravi) to India and the Western Rivers (Indus, Jhelum, Chenab) to Pakistan, while granting India specific non-consumptive hydroelectric rights on the Western tributaries.

However, climate-driven glacial retreat, shifting monsoons, and heightened political tensions have pushed the treaty toward structural collapse. India has formally requested modifications to the treaty, citing changing demographic needs, clean energy priorities, and persistent procedural delays caused by Pakistani legal challenges to Indian hydro-electric projects (such as Kishanganga). A collapse or unilateral departure from the IWT would jeopardize water and agricultural security for over 200 million people in Pakistan's agricultural heartland, transforming shared water resources into a flashpoint for conflict.

8. STRATEGIC RECOMMENDATIONS & GLOBAL SOLUTIONS

ACTIONABLE DIPLOMATIC FRAMEWORKS

- **Multilateral Atmospheric Oversight:** Establish a UN-backed governance regime and global moratorium on outdoor Solar Radiation Management deployment pending comprehensive risk assessment and universal consensus.
- **Transboundary Benefit-Sharing Models:** Transition transboundary river agreements away from rigid volumetric allocation toward shared economic benefits, joint flood management, and real-time data integration.
- **Equitable Adaptation Finance:** Enforce climate finance commitments from major historical emitters to bolster regional water resilience, ensuring vulnerable states do not rely on high-risk geoengineering measures.